

Characterizing User Behavior on a Mobile SMS-Based Chat Service

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Abstract. *The use of mobile instant messaging (IM) services has grown significantly last years. Usually, mobile chat services work over the Internet using cellphone carriers' resources, such as the SMS (Short Message Service) platforms. Understanding the user behavior in this environment is paramount to improve service performance and user experience. In this article, we present and discuss a characterization of the user behavior on a mobile SMS-based chat service. We describe the usage patterns of this service providing a daily perspective of user behavior. We show that a very small group of heavy users consumes a significant amount of carrier's resources. Moreover, we also present the transitions and navigation patterns of this very small group of users to understand their peculiar behavior.*

1. Introduction

Mobile instant messaging (IM) services have been outstanding as important communication tools by connecting an increasing number of persons at any time of the day at any place around the world. According to [Mander 2014], about 600 million adults are currently using IM services on their mobile devices provided by mobile applications like *Viber*, *Kik*, *WhatsApp*, *Line*, and *WeChat*. Usually, these applications work over Internet. Nevertheless, similar short message service (SMS) services based on the exchanging of short messages have been provided by cellphone companies around the world, such as Vodafone¹, Orange² and Safaricom³. Whereas the massive data volume generated by these services over networks' resources should be handled by mobile service providers, they need to understand the behavior of their users to improve user experience, performance, availability, cost, and quality of offered service.

The present article characterizes user behavior on a mobile SMS-based chat service provided by a major cellphone carrier in Brazil. Users pay a monthly flat rate to access a set of chat rooms provided by carrier. These rooms are organized by subjects to users send short messages to others with similar interest. They also can create private rooms to chat particularly with other users. In early 2014, about 335,000 messages per day were exchanged on this service. Considering that the service is not free and is based on SMS, this volume is enough expressive.

In particular, we provide an extensive analysis of the service's usage patterns considering a dataset composed by two million messages exchanged among more than 20

¹<http://www.vodafone.in>

²<http://www.orange.mu>

³<http://www.safaricom.co.ke>

thousand anonymized users throughout one week on May 2014. We identified different user profiles using the number of exchanged messages, the number of user sessions, and the frequency of messages exchanging as input to *X-means* clustering algorithm. In addition, we use the same features and clustering algorithm to provide a daily perspective of user behavior, thereby minimizing the effects of data aggregation. Furthermore, we present the transitions and navigation patterns considering the usage of service's rooms of a particular profile of *Heavy Users*, a very small group of users that send many messages. Moreover, we presented their navigational behavior using Customer Behavior Model Graphs (CBMGs) [Menascé et al. 1999].

The remaining of this article is organized as follows. Section 2 presents some related work which places our work in literature. In Section 3, we describe the dataset used to characterize user behavior on the mobile chat service. In Section 4, we present a comprehensive analysis on characterization results. Section 5 describes the usage behavior and the navigation patterns of particular user profiles. Finally, Section 6 points out the final remarks and a brief discussion on future work.

2. Related Work

There is a significant set of related works in literature towards characterizing IM services. Most of them focused on user behavior, particularly on users interactions in the workplace [Isaacs et al. 2002], message traffic and conversations [Zerfos et al. 2006], user engagement [Budak and Agrawal 2013], and service architecture [Fiadino et al. 2014]. Different from previous work in literature, we provide a characterization of a private SMS-based chat service to detect malicious or atypical user behavior.

[Xu and Wunsch 2005] show that clustering techniques has been applied in a wide variety of fields, ranging from life and medical sciences, engineering (machine learning, pattern recognition), computer sciences (web mining, spatial database analysis, data mining). In this article, we use the X-means algorithm [Hall et al. 2009], an extension to the K-means [Jain et al. 1999]. The both algorithms are commonly used in characterization works [Benevenuto et al. 2012, O'Donovan et al. 2013]. However, X-means provides improved functions, such as the automatic detection of the number of clusters to generate.

In [Lipinski-Harten and Tafarodi 2013], the authors argue that online users can act improperly since the negative impact of recrimination for inappropriate behavior is lower than in face to face communication. For example, users may not be inhibited from using offensive language or disclosure of inappropriate content, such as pornography and violence in chat rooms not suitable for such content. In this line, previous work in literature have proposed approaches to detect malicious behavior in online conversations [Frank et al. 2010, Gupta et al. 2012, Wollis 2011].

In addition to prevent malicious behavior, a major challenge for IM service providers is to improve service performance preserving user loyalty [Deng et al. 2010]. In this line, there are important aspects that must be considered, such as the size of the user neighborhood represented by the number of contacts of an user, and the degree of confidence and engagement of the user with the IM service. In [Zhou and Lu 2011], the authors argue that low cost, attractive features, and extreme competition are key factors for an user to migrate from one IM service to another.

In [Du et al. 2009], the authors suggest a model to investigate user behaviors

changing on weighted time-evolving networks, based on clique patterns and other features. Considering the user patterns, the authors detected suspicious behaviors in outliers – a particular group of users.

3. Dataset

The dataset used in our analysis contains messages exchanged on a mobile SMS-based chat service provided by a major cellphone company in Brazil⁴ during the week from May 10th to May 16th, 2014. The dataset includes 2,348,805 messages exchanged by 21,210 users who visited 34 different categories of chat rooms. The message exchanging occurs within 95,235 different sessions created by users. For privacy, user identifications were completely anonymized. Each record of the dataset represents one message sent by an user and contains the following fields:

- *Session Identifier*: an unique identifier of one user session; a new user session is created every time user initiates a navigation over the rooms of the mobile chat; after a downtime of 30 minutes, user session is finished.
- *Sender*: an unique identifier (anonymized) of the user that sent the message.
- *Category Identifier*: an unique identifier of the chat room category.
- *Category Name*: the name (label) of the chat room category.
- *Message*: the content of the message.
- *Message Type*: an unique identifier of the message type, i.e. *Private*, *Public*, and *Room*.
- *Timestamp*: sending message date and time.

The messages exchanged by users can be (i) *Public*, i.e. messages sent and accessible to all users in the chat room, (ii) *Room* messages sent to a single user but accessible by all users in the chat room, or (iii) *Private* messages sent to a single user and only accessible by this single user (one-to-one message).

The chat rooms are classified by their respective subjects, such as entertainment, sports, and cities, and by the nature of the content of their messages, such as restricted to 18 years old or elder. The *personal* class is used to identify chat rooms created by users. For analysis, we reorganized these chat room classes in categories as follows:

- *General*: messages of sports or religions.
- *Location*: messages related to cities and regions.
- *Person*: messages in personal chat rooms.
- *Relationship*: messages about nightlife or flirting.

4. Mobile Chat Service Overview

Different from other popular IM players such as *Viber*, *Kik*, *WhatsApp*, *Line*, and *WeChat*, which provide mobile applications with rich interfaces and a sort of facilities on the screen, the chat service considered in the present work is totally SMS-based. For instance, if a user is in a chat room and want to send a message to another user in the same chat room, the sender user must send the sequence of commands “*T + destination nickname + text message*”, where T is the abbreviation to *Talk*. There are a lot of another commands that vary according to the context in which the user is in the service, for example view the available categories, the rooms of a certain category, perform administrative actions such as changing the nickname among others. In addition, there is a significant user engagement, as the service has about 335,000 messages exchanged during one day.

⁴To avoid violate privacy policies, company name and dataset details will be preserved.

4.1. Messages by Categories

Figure 1 presents the message exchanging in the mobile chat service on a daily perspective. The messages are organized by chat rooms' categories. From Figure 1, we observe that the highest amount of messages exchanged in a day occurs on Wednesday, corresponding to 14,95% of all exchanged messages in the week. Additionally, the lowest amount of message exchanging in a day occurs on Sundays and Mondays.

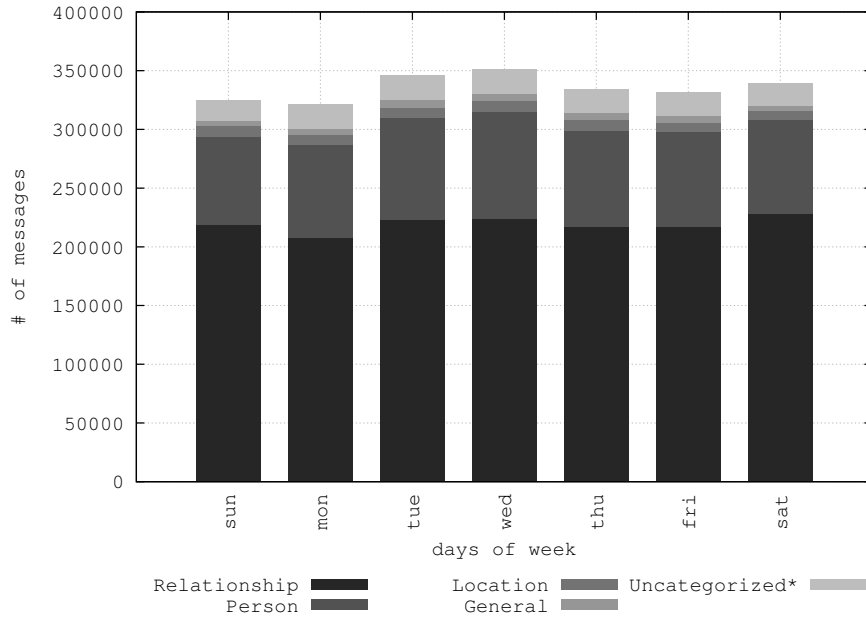


Figure 1. Messages exchanging by day and by category. *Uncategorized* messages refers to *Private* messages.

We can also observe from Figure 1 that Relationship messages correspond to 65% of all message exchanging during the week. Note that, 24% of messages are exchanged inside “*Person*” chat rooms, where users can talk about different subjects. Moreover, about 89% of all messages are exchanged in a small number of chat rooms without a specific subject.

Figure 2 presents the amount of exchanged messages over the hours of each day of the week. The darker area represents the greater amount of exchanged messages in each hour of the day. From Figure 2, we observe that highest peaks of usage occur commonly in the evenings, from 6pm to 10pm. In this time range, occurs about 36% of all message exchanging. During the afternoons, the amount of exchanged messages is also significant, corresponding to 26% of all messages. As expected, the message exchanging declines from 1am to 7am.

Nevertheless, the amount of messages exchanged per day does not vary significantly, what is very common in network traffic, but it does not occur in the SMS application. As this service creates opportunities to entertainment and social relationships, we believe the evening massive usage is related to a kind of “social need” of users. The non-occurrence of a weekly fluctuation and the high use of service in the evenings could be explained by this need, as we can observe from Figures 1 and 2.

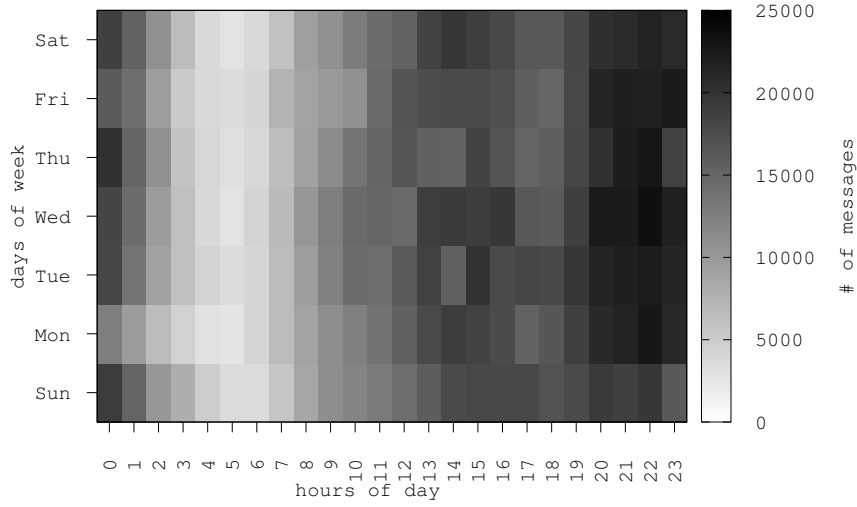


Figure 2. Message exchanging throughout the day

4.2. User Sessions and Message Types

In this section, we present two *Venn Diagrams* to represent the amount of sessions created by users and the number of messages of each category, respectively. The numbers on the labels represents the related field on the diagram. For example, from Figure 3, we observe that 45,049 user sessions contains exclusively room messages. We also observe that in 7,950 user sessions the three type of messages are present.

From Figure 3, we observe that in more than 87% of the user sessions we have exclusively *Public* and *Room* messages, suggesting a non-confidentiality pattern in the message exchanging. Moreover, almost half of user sessions are exclusively formed by *Room* messages, which suggests that users mostly communicate pairwise, but without worrying about the privacy of the communication.

Figure 4 shows that almost 77% of the messages are exchanged in non-confidential user sessions, i.e. user sessions where only *Public* or *Room* messages are exchanged. This “open communication” suggests user interest for new relationships. Additionally, more than 22% of messages are exchanged in non-exclusively confidential user sessions, while less than 1% of the messages are exchanged in private user sessions. Thus, many users build new relationships in non-confidential user sessions, and some of them intensify existing relationships in private user sessions, probably motivated by the communication context and mutual interest.

The recognition of communication context can help to characterize user behavior, since the message exchanging motivated by a specific interest follow regular patterns [Greenfield and Subrahmanyam 2003]. However, context recognition in non-confidential user sessions is a challenging problem, since many users are sending messages at the same time, frequently changing the conversation subject.

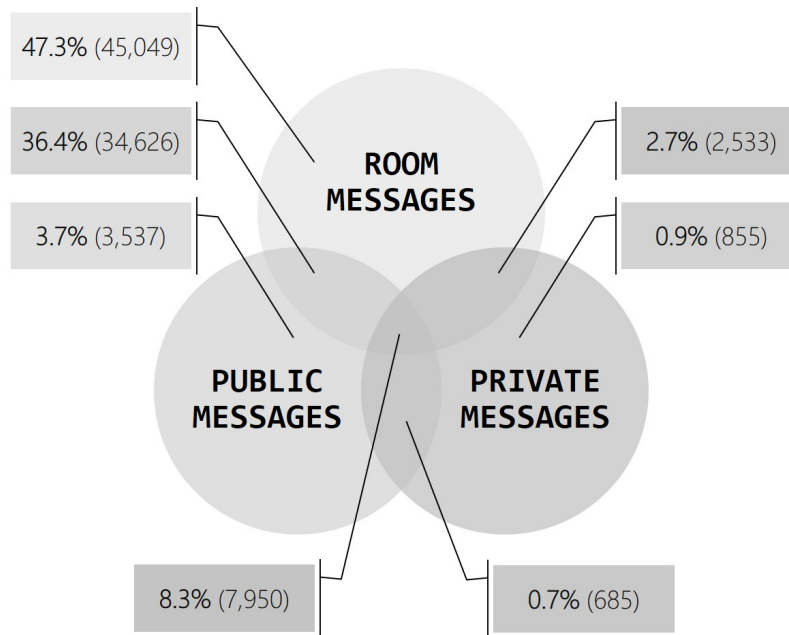


Figure 3. User sessions by message type

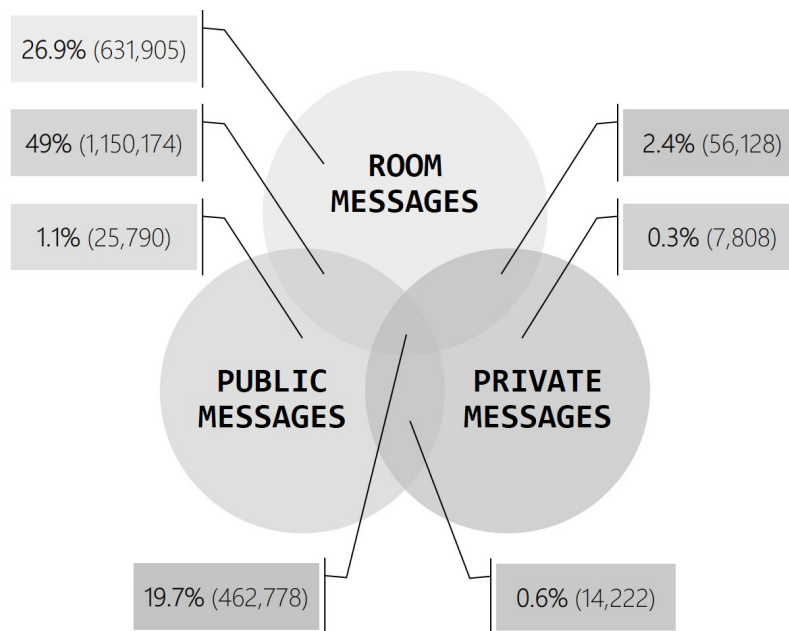


Figure 4. Messages by type on user sessions

5. User Behavior Analysis

We divide the user behavior analysis into three parts: (i) analyzing user message exchanging distribution; (ii) discovering user profiles using clustering techniques; and (ii) analyzing user transition and navigation patterns across chat rooms.

5.1. User Message Exchanging Distribution

In this section, we present the user message exchanging distribution in the mobile chat service. From Figure 5 we can observe that the user message exchanging behavior follows

a heavy-tailed distribution [Clauset et al. 2009], with a very small number of users sending the majority of the messages and the most of the users sending a very small number of messages on the chat service.

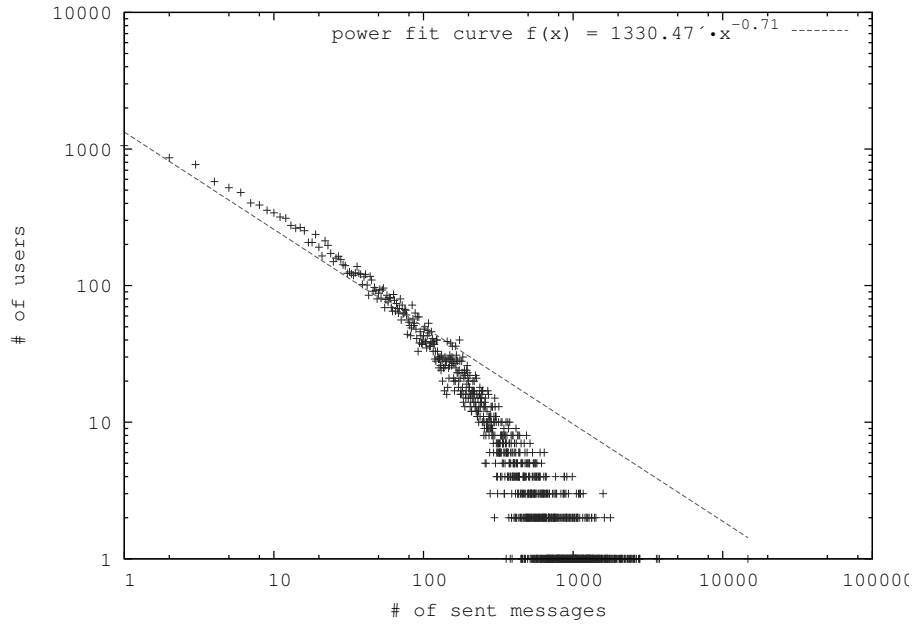


Figure 5. User message exchanging distribution.

Heavy-tailed distributions characterize an important number of behaviors from nature and human endeavor and have significant consequences for our understanding of natural and man-made phenomena. Particularly, in this article we show different user behavior on the chat service focusing our analysis on the head of the heavy-tail distribution, in a special and very small group of users which exchanges the majority of the messages.

5.2. Discovering User Profiles

In the following sections, we present a detailed characterization about user profiles who use the mobile chat service. We analyzed data in weekly and daily perspectives to understand user behavior.

5.2.1. Weekly Perspective

As aforementioned in Section 3, one user session is created every time an user initiates a navigation in the mobile chat service. Inside the session, the user exploits several chat service resources, such as listing available chat rooms by category and requesting support service. In this article, we only use the message exchanging service to discover user profiles, i.e., sets of users with similar behavior. Particularly, we consider three features about each user as input to the clustering algorithm which groups similar users:

- **Messages:** the number of exchanged messages.
- **Sessions:** the number of user sessions.
- **Frequency:** the rate of message creation per minute.

We use the X-means clustering algorithm [Pelleg et al. 2000] to discover user profiles. The X-means algorithm extends the popular K-means algorithm [Jain et al. 1999] by not only providing the clusters, but also estimating the suitable number of clusters should be created. These algorithms have been commonly used in clustering problems [Benevenuto et al. 2012, O’Donovan et al. 2013]. X-means creates clusters by minimizing the sum of the squared distances between each vector representing the averaged properties of each group and the cluster’s centroid. The distance between two vectors is computed by the Euclidean distance.

In this article, we use a well known implementation of the X-means algorithm [Hall et al. 2009] setting the maximum number of clusters to 10. Table 1 shows the four clusters provided by X-means in a weekly perspective, the percentage of users in each cluster, as well as the respective features (average values) for each cluster. In addition, it presents the coefficient of variation (CV , i.e. $\frac{Std.Dev.}{Average}$) for each feature to help understanding how cohesive is the cluster.

Table 1. Cluster’s overview in a weekly perspective

Cluster	Users	Messages		Sessions		Frequency	
	%	Avg	CV	Avg	CV	Avg	CV
Light	65.00	33.16	1.59	1.55	0.48	0.77	9.43
Infrequent	25.00	156.08	0.94	6.26	0.34	0.59	2.89
Frequent	8.00	440.59	0.86	16.62	0.24	0.57	0.58
Heavy	2.00	934.63	0.99	36.47	0.29	0.67	0.55

The first cluster contains 65% of all users. Users in this cluster exchanged few messages, approximately 33 per user session. The average frequency of message exchanging is almost 1, which is considered a high interaction frequency. However, users in this cluster typically access the service less than twice during the week. We named this user profile as *Light Users*.

About 25% of users are in the second cluster. Users in this cluster exchanged more messages than *Light Users*, approximately 156 per user session. The average frequency of message exchanging for this cluster is slightly lower, approximately 0.6. Users in this cluster typically access the service six times during the week. We named this user profile as *Infrequent Users*.

The users in the other two clusters exchanged several messages, using the service intensively. In the third cluster we have 8% of the users. Users in this cluster exchanged several messages and access the service about 20 times during the week. Due this behavior, we named this user profile as *Frequent Users*.

Finally, in the fourth cluster we have the remaining 2% of users which exchanged a high amount of messages. They access the service about 40 times during the week. We named this user profile as *Heavy Users*. This group represents only 2% of the users but exchanged about 14% of all messages and creates about 14% of all user sessions in the service. Due to this behavior, *Heavy Users* receive further attention in our analyzes.

5.2.2. Daily Perspective

We also use the X-means clustering algorithm and the same three features described in Section 5.2.1 to analyze the usage of the mobile chat service on a daily perspective. For comparison, we set the number of clusters to four, the same number of clusters found in the weekly perspective presented in Section 5.2.1, rather than allowing X-means to automatically discover the suitable number of clusters. Figure 6 presents the proportion of users in clusters in a daily perspective.

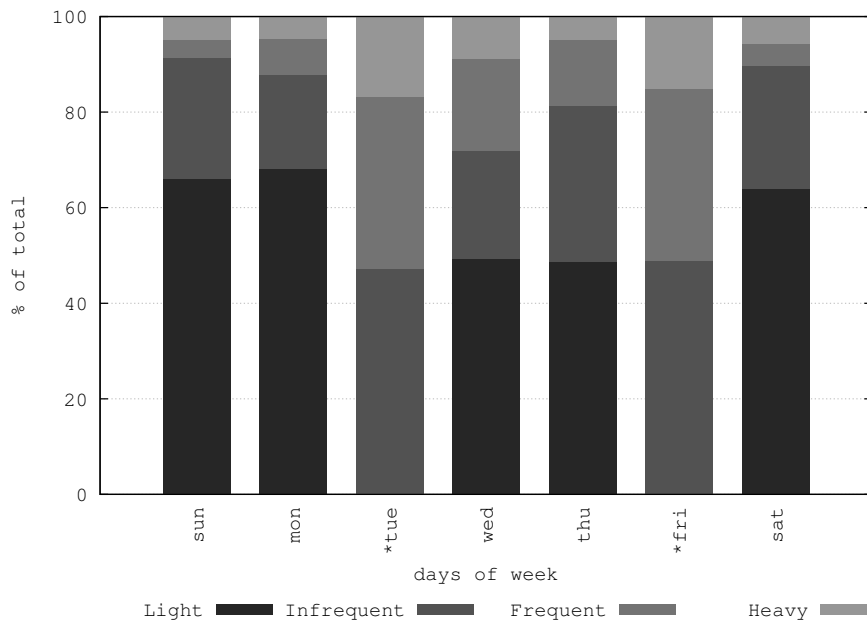


Figure 6. Proportion of users in clusters in a daily perspective.

From Figure 6, we observe that the proportion of users in clusters is similar to the weekly perspective, with a dominance of the *Light Users*, followed by *Infrequent Users*, *Frequent Users*, and *Heavy Users*. The exception occurs within two days of the week, Tuesday and Friday, when there is almost no *Light Users* using the service. In these cases, probably the *Light Users* have changed their behavior in the other days using the service more frequently.

Table 2 presents the four clusters provided by X-means in a daily perspective, as well as the respective features (average values) for each cluster. In addition, it presents the coefficient of variation (CV) for each feature.

Cluster	Messages		Sessions		Frequency	
	Avg	CV	Avg	CV	Avg	CV
Light	17.56	0.34	1.33	0.24	0.82	0.28
Infrequent	49.28	0.40	2.38	0.44	0.58	0.06
Frequent	112.41	0.39	4.41	0.55	0.60	0.11
Heavy	181.18	0.34	5.55	0.27	0.62	0.08

From Table 2 we observe that, similarly to the weekly perspective presented in Table 1, *Heavy Users* exchanged a high amount of messages per day, corresponding to almost 4 times more message exchanging than the *Infrequent Users* and 10 times more message exchanging than the *Light Users*, the two most representative groups. Additionally, *Heavy Users* created 3 times more user sessions than the *Infrequent Users* and 6 times more user sessions than the *Light Users*. Moreover, on a daily basis, the interaction frequency of the *Infrequent Users*, *Frequent Users*, and *Heavy Users* is almost the same. Since the average amount of exchanged messages by *Heavy Users* is significantly greater than the other groups, we conclude that *Heavy Users* use the message exchanging service for longer.

5.3. Transition and Navigation Patterns

As mentioned in Section 5.2.1, *Heavy Users* represent 2% of the users, exchanging about 14% of all messages and creating about 14% of all user sessions in the message exchanging service. In this section, we focus our analyses on *Heavy Users* investigating the user profile transition and navigation patterns of this peculiar user profile.

Particularly, to understand the user profile transitions, we identify *Heavy Users* in a day (D), recognizing their user profile in the day before ($D-1$). In addition, we analyse how *Heavy Users* back to the mobile chat service, recognizing their user profile in the day after ($D+1$). Table 3 presents the *Heavy Users* composition on a $D-1/D$ perspective. The D parameter was defined considering users with sessions between 0:00 and 23:59. By this, we were considering a daily perspective.

Table 3. Heavy Users composition on a $D-1/D$ perspective

Light	12.59%
Infrequent	21.91%
Frequent	20.06%
Heavy	30.99%
New Heavy Users	14.46%

From Table 3, we observe the majority of *Heavy Users*, almost 55%, in D belong to different user profile in $D-1$. In particular, almost 42% of *Heavy Users* in D were *Infrequent Users* or *Frequent Users* in $D-1$. Additionally, almost 13% of *Heavy Users* in D were *Light Users* in $D-1$. Moreover, the remaining 14% represents new *Heavy Users* that do not use the message exchanging service in $D-1$.

Table 4 presents the *Heavy Users* engagement on a $D/D+1$ perspective. From Table 4, we observe that more than 85% of *Heavy Users* in D back to the message exchanging service in the next day, and about 42% of them back with the same user profile. We can conclude that *Heavy Users* tend to remain in this behavior, since almost 31% of the users in this profile were already *Heavy Users* in $D-1$.

This group of *Engaged Users* that remain *Heavy Users* over time frequently returning to the service contribute to reinforce the *Heavy Users* behavior intensively exploiting service resources.

To understand the navigation behavior of *Heavy Users*, we use a *Customer Behavior Model Graph* (CBMG), a state transition graph that has been used to describe the

Table 4. Heavy Users engagement on D/D+1 perspective

Return rate	85.18%
Light	13.21%
Infrequent	17.64%
Frequent	26.92%
Heavy	42.22%

navigation patterns of groups of users [Menascé et al. 1999]. In this graph, each edge represents a transition probability from one node to another and each node represents a possible state to reach. Figure 7 presents a CBMG of the transition behavior for user profiles in a daily perspective. In this graph, each node represents one user profile and each edge represents the transition probability between user profiles. In addition, we also represent two abstract nodes in the graph, representing the start (*entry*) and the end (*exit*) states. We also highlight the paths with the highest transition probabilities.

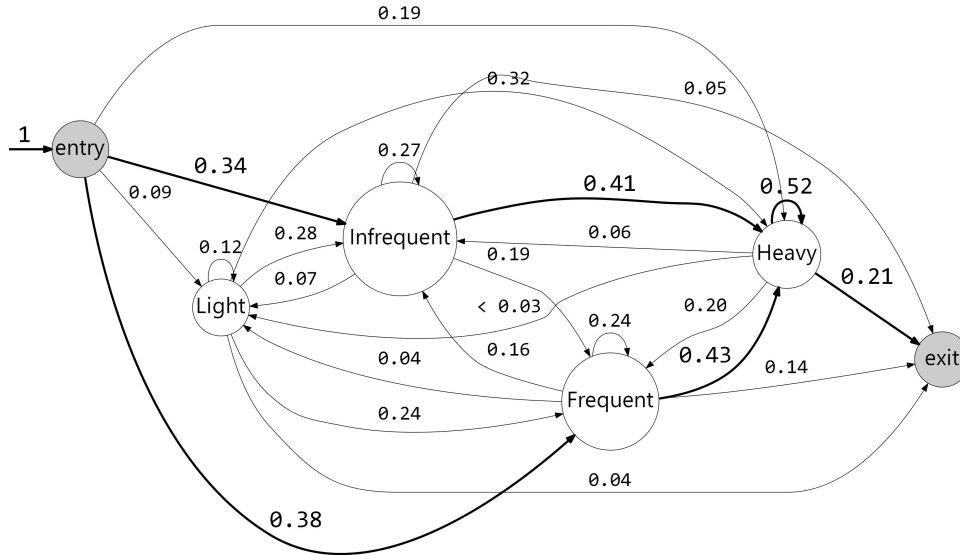


Figure 7. CBMGs for behavioral changes. The paths with the highest probability were highlighted.

From Figure 7, we observe that the *Heavy Users* change their behavior during the week. They are more likely to be initially classified as *Frequent Users*, with a probability of 0.38, followed by *Infrequent Users*, with a probability of 0.34. In both cases, users that are classified in these behavior have a high tendency to migrate to the group of *Heavy Users*, with an average probability of 0.42, remaining until the end of the period with a probability of 0.52.

Figure 8 presents a CBMG of the chat rooms exploitation by category in a daily perspective. In this graph, each node represents one chat room category and each edge represents the transition probability between chat room categories. Additionally, we also represent the abstract nodes *entry* and *exit* in the graph, and we also highlight the paths with the highest transition probabilities.

From Figure 8, we observe that *Heavy Users* usually start a session in the chat

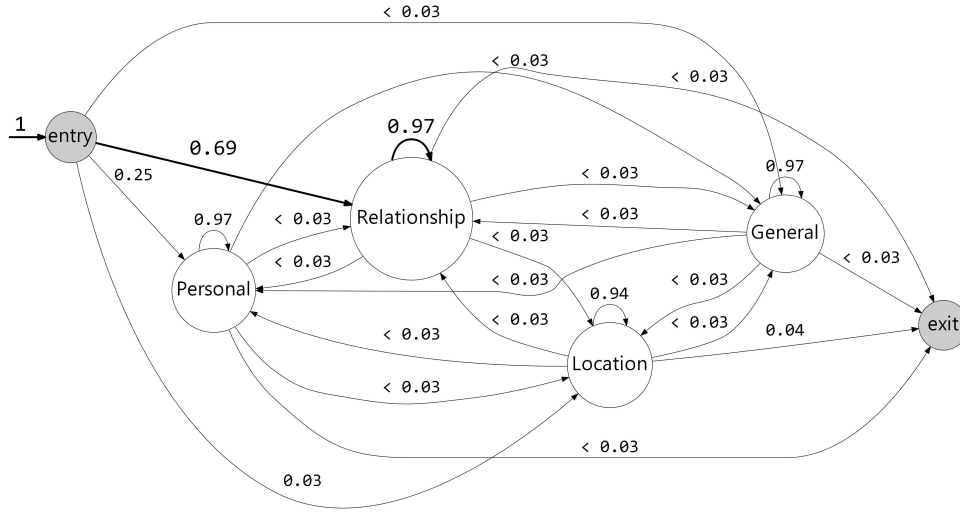


Figure 8. CBMGs for categories exploitation. The paths with the highest probability were highlighted.

through a room from the Relationship category, with a probability of 0.69. Once in a room from this category, the *Heavy Users* have an extremely high chance of staying in this type of room, with a probability of 0.97. The transitions from this state have little significant values, showing that *Heavy Users* effectively look for rooms of type Relationship.

6. Conclusions and Future Work

In this article we presented a comprehensive characterization of the user behavior on a mobile SMS-based chat service provided by a major cellphone company in Brazil. In particular, we described the usage patterns of this service using a dataset with millions of short text messages exchanged between thousands of users during a week.

In this high traffic IM service, message exchanging occurs mostly in the afternoons and evenings, in the middle of the week and inside *Relationship* chat rooms, with the majority of messages being accessible by anyone inside a chat room. Additionally, the weekly and daily perspectives of the user behavior points to the existence of four distinct groups of users: i) a large group of *Light Users* (65%) that exchanges very few messages with a very small gap between message exchanging and uses the service less than two times a week; ii) a group of *Infrequent Users* (25%) that exchanges few messages with a small gap between message exchanging and return to the service constantly; iii) a small group of *Frequent Users* (8%) that uses the service three times more frequently and exchanges more messages than *Infrequent Users*; iv) a very small group of *Heavy Users* that uses the service two times more frequently and exchanges much more messages than *Frequent Users*.

By focusing our analysis on the transition and navigation patterns of this very small group of *Heavy Users*, we show that these users tend to keep their behavior over time. In addition, they are engaged users that frequently back to the service intensively exploiting its resources. Moreover, we show that a significant part of *Infrequent Users* and *Frequent Users* change their behavior becoming *Heavy Users*. Analyzing the chat category exploitation, we show that *Heavy Users* look for Relationship chat rooms and

stay there.

The behavior patterns aforementioned about the *Heavy Users*, such as the amount of exchanged messages, the number of created user sessions, and the high service engagement, suggest be likely to find in this very small group of users those with a potential malicious behavior. Considering possible directions for future research, directly inspired by or stemming from the results of this work, we plan to investigate the message content of the *Heavy Users* to detect malicious behavior, such as defamation, pedophilia, phishing, and spamming.

We also plan to use other clustering algorithms and investigate different features, such as the distribution of messages by category, the duration of user sessions, and the message content. Another direction is to cluster user behaviors instead of users, looking for behavioral classes such as exploring and flirting. There are some techniques designed to capture roles and their dynamics, as suggested in [Fu et al. 2009, Nasraoui et al. 2008].

Moreover, we plan to further investigate transitions evolving private messages. As we observed, less than 1% of the messages are exchanged in private user sessions, suggesting that the final goal of the users is to get the contact number (e.g Whatsapp or another private way of contact) of the person, so they will be able to chat in a more friendly environment, away from any possibility of moderation. Once they do it, they will stop using the private chat (and the chat itself).

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